

Tezpur University
Spring Semester End Examination - 2025

Course Code: CO503

Course Title: Fuzzy Logic and Neural Networks

Full marks = 50

Time = 2 hours

(The figures in the right-hand margin indicate full marks for the questions)

1. State true or false:

1 × 5 = 5

- (a) Each alpha-cut of a fuzzy equivalence relation is a crisp relation.
- (b) Presence of strong modifier decreases the truth value of the associated proposition.
- (c) $P(X, Y) \circ Q(Y, Z) = Q(Y, Z) \circ P(X, Y)$, where $\circ$ indicates composition of fuzzy relations.
- (d) Perceptron rule updates a weight only if error occurs.
- (e) A 2-layer neural network can be trained using the Hebb rule.

2. Give one example for each of the following:

1 × 5 = 5

- (a) Fuzzy quantifier.
- (b) Truth values of a 5-valued standard Lukasiewicz logic.
- (c) Graphical representation of a binary fuzzy relation on a single set.
- (d) Reflexive and symmetric fuzzy relation in the membership matrix form.
- (e) Graphical representation of activation function used in delta training rule.

3. (a) Given the two sets $U = \{a, b, c\}$ and $V = \{x, y\}$. What does the following fuzzy set represent? 1

$$B = \frac{0.3}{(a,x)} + \frac{0.7}{(a,y)} + \frac{1.0}{(b,x)} + \frac{0.7}{(c,y)}$$

(b) Fuzzy sets representing heavy and light rainfall (in mm/hour) are defined as follows:

$$\text{heavy} = \frac{0.3}{2} + \frac{0.4}{8} + \frac{0.6}{12} + \frac{1}{30}$$

$$\text{light} = \frac{0}{30} + \frac{0.1}{20} + \frac{0.5}{12} + \frac{0.8}{8} + \frac{1}{5}$$

Develop fuzzy sets representing rainfall described by the following linguistic phrases:

- i) very heavy ii) fairly heavy iii) not very light

3

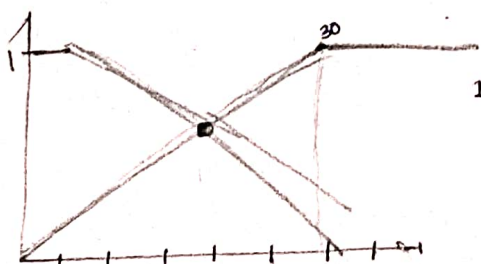
(c) Prepare a truth table showing the following logic operations: $a \wedge b$, $a \vee b$, $a \rightarrow b$ for 3-valued logic with truth values $\{0, \frac{1}{2}, 1\}$. Use Lukasiewicz implication: $a \rightarrow b = \max(b, 1-a)$. 3(d) Evaluate the logic expressions: $a \wedge b \rightarrow c$ when $a = 1$ and $b = \frac{1}{2}$ and $c = 0$. 1

4. (a) Why do you need defuzzification in a fuzzy processing system. 2

(b) Describe the center of gravity defuzzification method for discrete membership functions. 2

(c) Defuzzify the following fuzzy set using the center of gravity method. 2

$$A = \frac{0.1}{1} + \frac{0.3}{2} + \frac{0.8}{5} + \frac{0.6}{7} + \frac{0.2}{10}$$



P.T.O.

5. (a) Given the fuzzy sets A, B, C find the relations $R(A, B), S(B, C)$ and $R \circ S$. 3
- $$A = \frac{0.3}{a} + \frac{0.5}{b} + \frac{0.8}{c} \quad B = \frac{0.7}{a} + \frac{0.4}{b} \quad C = \frac{0.2}{1} + \frac{0.6}{2} + \frac{0.1}{3}$$
- (b) Represent the computed relation T in two other forms. 2
6. Consider the fuzzy rule "IF the temperature is high THEN the humidity is low".
- (a) Present the general structure of the fuzzy rule given above. 1
- (b) Name the linguistic variables involved in the fuzzy rule. 1
- (c) Provide your definition of the linguistic variables by graphically representing their possible states. 2
7. (a) Present an architectural diagram of a simple neural network for 2-class classification problem considering 3-dimensional patterns as input. Include bias weights at the processing nodes. 1
- (b) Explain why do we use a bias weight at the processing nodes of a neural network. 2
- (c) What is a training set and how is it used to train neural networks? 2
- (d) Present the Hebb learning rule in mathematical form. 1
- (e) What are the disadvantages of the Hebb learning rule? 1
- (f) Apply Hebb rule to find the weights required to perform the 2-class classification shown in the Table-1. Consider the neural network architecture presented by you in question (a) above. Show the final weight values in the architectural diagram. 4
- (g) Test the network using the first two patterns of Table-1. Use bipolar step function with threshold zero as the output unit's activation function. 2

Table-1

Input	Target class
(-1, -1, -1)	-1
(-1, -1, 1)	1
(-1, 1, -1)	-1
(-1, 1, 1)	1
(1, -1, -1)	-1
(1, -1, 1)	1
(1, 1, -1)	-1
(1, 1, 1)	1

8. Justify the following statements: 2×2=4
- (a) Binary representation of data is more advantageous as compared to bipolar representation.
- (b) Perceptron learning rule is more powerful than Hebb learning rule.

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